

Florida Atlantic University
Department of Engineering and Computer Science
EGN 4952C-001: RI: Engineering Design 2
Spring 2026

Decentralized Storage Network for Online Identification

Members:

Kanish Dangol

kdangol2023@fau.edu

Syed Shah

syedshah2023@fau.edu

Shiqiang Mo

mos2023@fau.edu

Xuan Gia Bao Nguyen.

xnguyen2023@fau.edu

Santiago Delgado

sdelgado2024@fau.edu

Mentor:

Mehrdad Nojournian

mnojournian@fau.edu

Faculty Advisor:

Washeem Ashgar

wasghar@fau.edu

Summary: Current online identity verification methods require interanauts to share sensitive information or documents to online service providers in order to gain access to a platform or service. While effective, there is a violation of privacy and risk of exposure if not protected correctly, and due to growing incidents of centralized data breaches, users show discomfort and lack of trust towards these procedures. This project aims to design and apply an alternative, decentralized method for online identity verification, prioritizing user's privacy.

Table of Contents

I.	INTRODUCTION	4
A.	Background - Shiqiang Mo	4
B.	Statement of the Problem	4
II.	SCOPE OF WORK	4
A.	Overview – Santiago Delgado	4
B.	Literature Review – Santiago Delgado	5
C.	Alternative Solutions – Santiago Delgado	8
D.	Evaluation – Kanish Dangol	8
E.	Decision – Kanish Dangol	8
III.	IMPLEMENTATION DETAILS	9
A.	System Specifications and Functionalities – Santiago Delgado	9
1.	Web Portal	9
2.	Admin Node	10
3.	Storage Network	10
B.	Overall System Design with Block Diagrams – Santiago Delgado	11
C.	Circuit diagrams, Flowcharts and/or Use Case Diagrams – Santiago Delgado	12
D.	Technology and Technical Standards used – Bao Nguyen	13
E.	Testing and Performance Evaluation – Santiago Delgado	14
1.	Information Upload/Update Testing	14
2.	Verification Process Testing	15
F.	Discussions on Lessons Learned – Santiago Delgado	18
IV.	CONCLUSION – Santiago Delgado	19
V.	REFERENCES	21

List of Figures

Figure 1: Decentralized ID Network System Interfaces 12

Figure 2: Key Fragmentation Diagram 13

Figure 3: Verification Process 13

List of Tables

Table 1: Data user inputs testing 15

Table 2: Verification requests testing 18

Table 3: Testing Summary..... 18

I. INTRODUCTION

A. Background - Shiqiang Mo

Online service increasingly requires users to verify identity-related information before accessing platforms, including age-restricted content, financial services, employment documentation, and the issuance of digital credentials, in many real deployments, verification is not simply an account feature but a core trust decision that determines whether a user is eligible to proceed. As a result, verification mechanisms must support both reliable assurance and user experience that people are willing to complete in practice.

Most current systems rely on centralized identity verification workflows that ask users to provide sensitive personal information such as government IDs, biometric scans, or full personal records, while these methods can be effective, they often require users to share more information than necessary, creating privacy concerns and discomfort, for example, proving a simple attribute (e.g., being over 18) can unintentionally expose unrelated personal data such as a home address or identification number. In addition, centralized storage of identity documents introduces security risks: a large database of personal information is a valuable target, and breaches demonstrate the vulnerability of centralized systems. Once compromised, identity information cannot be reset like a password, increasing long-term consequences for users. At the same time, lightweight methods such as self-declared age entries are weak and easily bypassed, making them ineffective for meaningful identity assurance.

B. Statement of the Problem

The core problem is how to verify identity information securely without requiring full disclosure of personal data and without depending on a single centralized database. Specifically, the system should allow a verifier to confirm a required attribute (e.g., identity or age eligibility) using minimal outputs such as a “yes/no” decision, while preserving user privacy and reducing exposure to large-scale compromise. The solution must balance (1) security and resistance to trivial bypass, (2) privacy preservation and data minimization, and (3) practical implement ability within the constraints of user secure decentralized network, including realistic deployment and operational requirements.

II. SCOPE OF WORK

A. Overview – Santiago Delgado

We aim to design a proof-of-concept prototype for an alternative zero-knowledge–based approach to online identity verification. The system will consist of an automated peer-to-peer network of nodes managed by a trusted authority, responsible for securely storing, protecting, and validating user data. Unlike traditional methods, this approach requires only a single in-person verification with the trusted authority. Afterward, third parties can submit verification requests and receive only a yes/no response, without accessing the underlying personal data.

To demonstrate this model, we will simulate the complete process through a web portal that enables interaction for both users and verifiers. Users will upload their information to the network, where it will be encrypted and secured through key fragmentation. Verifiers will be able to submit requests specifying the attributes they need to confirm (e.g., minimum age requirement). An administrative server node will act as the interface between users, verifiers, and the decentralized network. This node will function as an API server, maintain a SQL database, and serve as the network’s entry point.

Our goal is to replicate a realistic deployment scenario by running interconnected nodes on separate machines, hosting the web portal and administrative server locally, and enabling live interaction with the system to validate the feasibility of the proposed solution.

B. Literature Review – Santiago Delgado

To better understand and appreciate the impact of our project, we must look at the current state of the art of the issues and technologies we address. The problem we aim to solve is one that has already been discussed many times before, researchers actively try to design and implement new ways to verify identity without violating online anonymity and privacy. In this brief overview we will go over the relevance of this issue in the cybersecurity field, and the proposed frameworks and how they relate to our approach.

With the proliferation of online services (OSs) and social media platforms (SMPs) and their growing userbase, it has become crucial to authenticate and manage permissions with the implementation of secure, accurate access control mechanisms. Public online spaces have become a nest of mature, sensitive content that is certainly not suitable for all audiences. The main concern is to keep kids outside of these spaces by implementing age verification (Forland et. al, 2024). The process by which this verification is done varies by type, which directly affects the effectiveness and accuracy of the age check (Jarvie & Renaud, 2021; Forland et al., 2024; Eltaher et al., 2025).

Other situations in which strict authentication might be required are , as explored by Satybaldy et al. (2022) with their self-sovereign identity technology framework for document verification. Purposes such as verifying verifiable credentials (VCs) for taxes, employment, or digital identifications are mentioned. Many other authors also focus on the verification of identity, as it is the most useful application when it comes

to authentication processes (Jamal et al., n.d; Deng et al., 2024; Takeuchi et al., 2024; Tyagi et al, 2018).

Recent studies have examined the range of age-assurance and verification mechanisms used by major SMPs and OSs. Eltaher et al. (2025) assess the different types of age assurance and verification methods provided in major SMPs and OSs by going through the registration process with new accounts and observing the required verification steps. This provided a firsthand experience of how these online spaces verify age and their resilience to bypassing attempts. Similarly, Jarvie and Renaud (2021) experimented with current age verification and assurance mechanisms to test different ways to bypass it. They highlight that verification practices range widely, from minimal “verification theatre,” such as clicking a checkbox, to highly privacy-invasive identity checks. A similar analysis is provided by Marsden (2023), who explores the legal framework regarding age verification and privacy protection online. Collectively, these studies show that platforms employ a spectrum of approaches, but most rely on either easily falsified self-reporting mechanisms or intrusive identity-based systems with substantial privacy implications.

There seems to be a consensus when it comes to the effectiveness of the current available mechanisms for online identity verification, including but not limited to age checks. Following the IEEE standard, it was found that major SMPs use Asserted age verification mechanisms, which is considered inadequate (Eltaher et al., 2025). They allow the users to easily bypass age restrictions by providing false information; or in other cases, trick verification softwares such as face recognition for age estimation, as tested by Jarvie and Renaud (2021). To properly verify the age and identity, SMPs and OSs often implement privacy-invasive mechanisms that could endanger the user’s online anonymity or violate their First Amendment (Marsden , 2023). It is imperative to find the balance between strictness and privacy-respecting procedures, which cannot be provided with the most common, available mechanisms offered today.

For years, researchers have considered the use of decentralized identity systems as a way to solve the privacy-invasive identity verification methods while keeping the information truthful and accurate. Self-Severeign Identity (SSI) systems have been explored as possible solutions, allowing users themselves to manage the use of their information during verification processes (Dib, O., & Toumi, K, 2020; Bachmann et al., 2019). With the implementation of blockchain and decentralized ledgers, decentralized identifiers (DID), digital wallets, and peer-to-peer networks, authors have come to design new schemes for online age and identity verification (Deng et al., 2024; Jamal et al., n.d; Takeuchi et al., 2023). The key concept of utilizing decentralization for DID management is the privilege of control, which shifts from an issuer-to-verifier process to a user-to-verifier, allowing users to better control the way they share their identifiable information (Satybaldy et al., 2022).

Deng, et al. (2024) introduce a new concept labeled as FutureDID, a combination of multiparty verification system that eliminates the need of a central authority. The system is built upon a blockchain platform that acts as a registry, in which users

register in. Then both issuers and verifiers can check the registry to either confirm/verify that the information is correct.

A similar approach is done by Jamal et al. (n.d) with their blockchain-based identity verification system. Users register to a blockchain network with the help of an authority entity, using the tamper-proof properties of the blockchain ledger to “print” it permanently. A verifier then can register as well and request to access the information during the verification process.

Takeuchi et al. (2023) focused their effort into utilizing a similar design, implementing blockchain as well, to prevent exfiltration ransomware attacks. With the addition of smart contracts to manage control access and encryption techniques to protect identifiable information, their solution promises to harden the system and provide a more trustworthy environment. Testing results show an improvement in cybersecurity attacks detection and response time.

Satybaldy et al. (2022) makes use of a blockchain ledger to preserve the identity of users, which then can manage access to their information using digital wallets. Their approach prioritizes making the SSI concept efficient, interoperable, and privacy-preserving. The concept relies on trusting institutions to register validated information in the ledger, to then give the users the means to manage access to it.

Future research direction focuses on the active innovation for new effective, non-invasive verification mechanisms that respect the balance between security and privacy (Jarvie & Renaud, 2021; Marsden, 2023; Forland et al., 2024; Eltaher et al., 2025). The implementation of SSI systems and the use of DID seems to be promising, but further research regarding optimization, testing, scalability capability, legal adaptation, and use of zero-knowledge proofs remain to be explored (Jamal, et al., n.d; Deng et al., 2024; Takeuchi et al., 2023; Alizadeh et al., 2022; Dib & Toumi, 2020; Bachmann et al., 2019).

The literature demonstrates both the urgency and the complexity of developing identity and age-verification mechanisms that are effective yet respectful of user privacy. Current approaches deployed by major online services are either too weak to prevent circumvention or too intrusive to be ethically or legally sustainable. As a result, researchers increasingly converge on decentralized, privacy-preserving models—particularly SSI and DID-based frameworks—as promising alternatives capable of shifting control back to users while maintaining trustworthy verification processes. However, despite meaningful progress, these solutions remain in early stages and face challenges in performance, scalability, interoperability, and regulatory adoption. The gaps identified across the literature highlight a clear research trajectory: designing verification systems that achieve robust security without compromising anonymity, and integrating emerging technologies such as blockchain and zero-knowledge proofs to support this balance. Our project positions itself within this evolving landscape, contributing to the ongoing pursuit of secure, accurate, and privacy-conscious online verification.

C. Alternative Solutions – Santiago Delgado

Current alternative solutions to online age and identity verification generally fall into two main categories: centralized identity-based systems and decentralized self-sovereign identity (SSI) frameworks. Traditional centralized approaches, commonly used by social media platforms and online services, rely on either self-declared information (such as entering a birthdate or clicking a checkbox) or document-based verification methods that require users to upload government-issued IDs or undergo biometric checks. While the former are easily bypassed and largely ineffective, the latter provide stronger assurance but raise significant privacy and anonymity concerns, often requiring users to disclose more personal information than necessary.

In response to these limitations, researchers have proposed decentralized identity models built on blockchain and distributed ledger technologies. These systems, often based on decentralized identifiers (DIDs) and verifiable credentials (VCs), shift control from centralized authorities to users through digital wallets and peer-to-peer verification. Frameworks such as FutureDID and other blockchain-based verification schemes aim to provide tamper-resistant, transparent, and privacy-preserving authentication mechanisms. Some approaches also integrate smart contracts, encryption, and emerging techniques like zero-knowledge proofs to allow selective disclosure of information (e.g., proving age eligibility without revealing exact birthdate). Although promising, these decentralized solutions still face challenges related to scalability, interoperability, regulatory compliance, and real-world adoption.

D. Evaluation – Kanish Dangol

We evaluated the different solutions based on security, privacy, scalability, and usability. Traditional centralized systems are either too weak or too invasive. Simple methods like entering a birthdate are easy to bypass. Stronger methods, such as uploading a government ID, improve security but require users to share sensitive personal information. This creates privacy risks and makes centralized databases attractive targets for hackers.

Decentralized self-sovereign identity systems improve privacy by giving users more control over their data. Many use blockchain and digital wallets, and some include zero-knowledge proofs to confirm information without revealing full details. However, fully decentralized systems can be complex, expensive, and difficult to scale.

Our proposed decentralized storage network combines the strengths of both approaches. A trusted authority performs a one-time in-person verification, and afterward, identity checks return only a yes/no response. This reduces data exposure while maintaining strong security and practicality.

E. Decision – Kanish Dangol

After evaluating the available alternatives, we decided to move forward with the decentralized storage network model managed by a trusted authority. We chose this approach because it provides a strong balance between privacy, security, and practicality. Fully centralized systems are too invasive and create single points of failure. Fully decentralized blockchain systems, while promising, are still complex and may be difficult to implement and scale within the scope of our senior design project.

Our design keeps the initial identity verification process secure by requiring a one-time, in-person check with a trusted authority. After that, the system allows verifiers to confirm identity information through a secure peer-to-peer network without directly accessing sensitive personal data. Instead of sharing full documents or personal details, the system only provides a confirmation response (such as “yes” or “no”).

This decision aligns with our project goals, to create a proof of concept system that demonstrates how identity verification can be secure, privacy oriented, and decentralized. By combining trusted verification with distributed storage, our solution contributes to ongoing research in privacy preserving digital identity systems while remaining realistic and achievable within our project timeline.

III. IMPLEMENTATION DETAILS

A. System Specifications and Functionalities – Santiago Delgado

1. Web Portal

Specifications

Built using:

- Node.js 16.1.6
- Tailwind CSS

Functionalities

The web portal serves as the front end of our system, the interface by which the users can interact with the service the network provides. Its main functionalities include:

- Provide good user experience through a friendly, easy interface.
- Manage user’s sessions, offering register/log in functions

- Allow controlled actions to the network, such as request verification of another user or upload information

2. Admin Node

Specifications

Built using:

- Express.js
- Libp2p Typescript/JavaScript implementation

Functionalities

The admin node works as a bridge between the web portal and the network, handling all the backend operations and logic. Main functionalities include:

- Interact with the database
- Handle API requests
- Contact network nodes

3. Storage Network

Specifications

Built using:

- Libp2p Go Implementation

Functionalities

The storage network is where the user information will be processed and stored, made of numerous automated individual nodes that communicate and collaborate to verify the information upon request. Main functionalities include:

- Receive user data, encrypt it, and store it
- Split encryption keys and distribute the fragments
- Gather fragments, decrypt data, and verify it against requested criteria

4. Databases

Specifications

We use two databases in this project:

- PostgreSQL – For web portal-related operations
- MongoDB – For storage nodes

Functionalities

The PostgreSQL database is to store user account information and verification requests entries.

The MongoDB database is to be installed in every network node to store the encrypted user information and key fragments.

B. Overall System Design with Block Diagrams – Santiago Delgado

In this system, there are three main entities interacting together: The admin, the user, and the service provider. Each one fulfills a necessary action in order to make the system work.

Admin – Is in charge of the network of nodes and the web portal, responsible for their maintenance. This entity also is responsible for verifying the information users submit to the network, only once each time it is updated. This is to make sure all the information saved in the databases is legally correct.

Service Providers – This entity will mainly use the system to make verifications requests to users, specifying what information it is to be verified and the user that needs the verification. This entity would correspond to companies or any other online service provider.

User – Common internet user, recipient of online services, that is the target of mandatory ID verification processes. They can use the web portal to upload their information and accept/deny requests from the service providers.

At the center of it all lays the decentralized identification network, which has been previously described in this document.

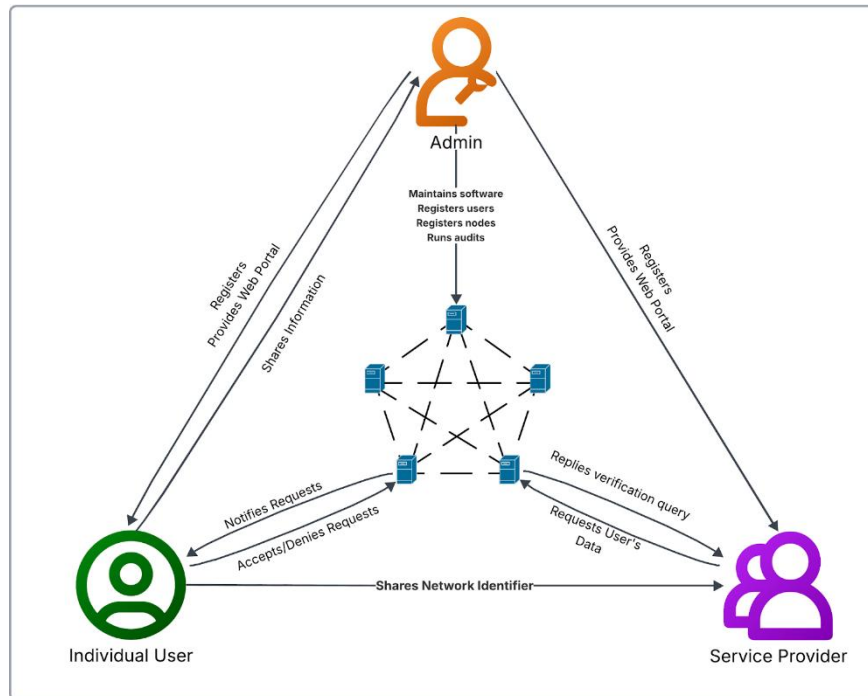


Figure 1: Decentralized ID Network System Interfaces

C. Circuit diagrams, Flowcharts and/or Use Case Diagrams – Santiago Delgado

The following diagram depicts a high level overview of how encrypted data and its key are treated by our system. Once the data is encrypted with a random key, the latter gets fragmented using Shamir's Secret Sharing. The key is divided into a certain number of shares, then each one of them is distributed randomly across the network. This way, not a single node can access the information, and the encrypted data may only be accessed when the user requests the nodes to put the fragments back together again.

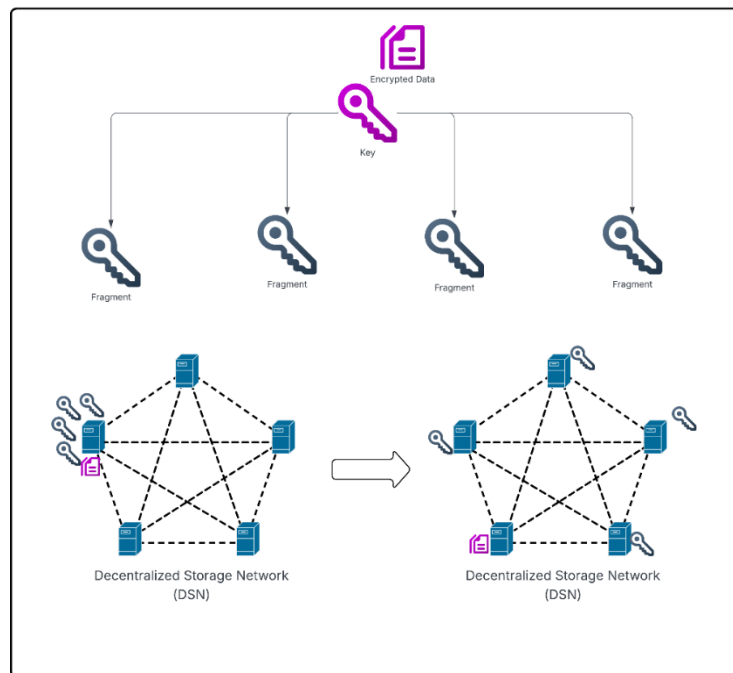


Figure 2: Key Fragmentation Diagram

The following diagram shows the practical real use case scenario our system is designed for. It starts with the user needing to access a service that requires verification. Using the web portal, and assuming the user's information has previously been submitted and verified by the system administrator, the respective service provider can submit a verification request with the user as the target. The user can review and choose to either accept or deny the request. Upon acceptance, the full request will be sent to the network, which will go through the automated verification process, and reply with a yes or no answer.

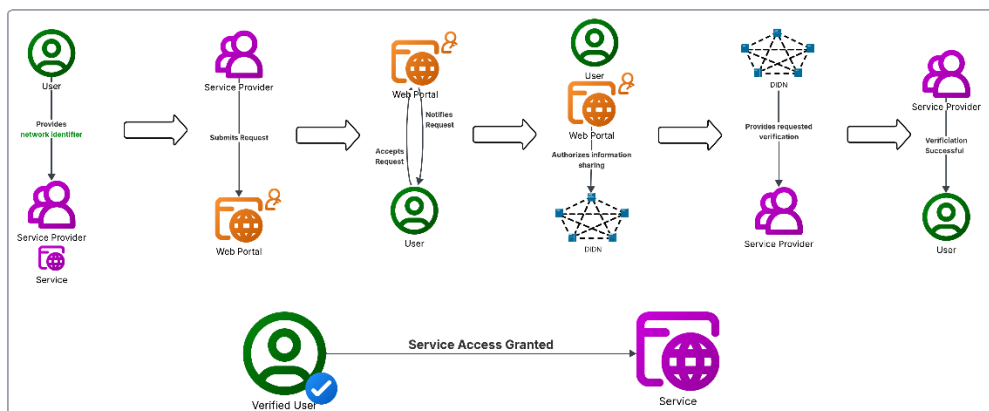


Figure 3: Verification Process

D. Technology and Technical Standards used – Bao Nguyen

1. Backend Technology:
 - Node.js: used for building web portal backend and handling server-side operations. Its event-driven architecture allows for efficient processing of multiple concurrent requests.
 - Express.js: used for managing communication between the web portal and the network. Its flexible framework enables efficient communication between the system components.
2. Frontend Technology:
 - Tailwind: used for designing a responsive and intuitive website interface due to its utility-first framework, which allows rapid UI development and consistent component styling.
3. Database Technology:
 - MongoDB: stores users encrypted data block, fragmented keys. Its document-oriented design allows efficient handling of unstructured, encrypted data,
 - PostgreSQL: stores users and service providers' login credentials and request information. The relational structure can ensure data integrity and consistency.
4. Network Technology:
 - Libp2p: used for establishing the communication layer between storage nodes on the network. Its modular and peer-to-peer architecture fits well with the core of the project, while also allows efficient interaction between nodes.
5. Cryptographic Technological Standard:
 - AES-256: used to encrypt user data before distributing to the network due to it being the industry-standard encryption algorithm, ensuring data security when stored on different nodes.

E. Testing and Performance Evaluation – Santiago Delgado

1. Information Upload/Update Testing

This test is to validate the inputs that the form in the web portal can take to submit their information to the network.

Test #	Inputs	Expectation	Results	Pass/Fail
T1	All field empty	Unable to submit	Unable to submit	Pass
T2	Name: Santiago Delgado Gender: Male Address: Fake Address 123, WPB, FL DOB: 2003-09-02	Data is successfully submitted Data reaches the nodes, it's encrypted, and finally fragments across nodes.	Data is successfully submitted Data reaches the nodes, it's encrypted, and finally fragments across nodes.	Pass
T3	Name: (empty) Gender: Male Address: 123 Main	Unable to submit	Unable to submit	Pass

	St, WPB, FL DOB: 2001-05-10			
T4	Name: Laura Chen Gender: Female Address: 987 Lake Rd, Orlando, FL DOB: 2027-01-01	Unable to submit	Unable to submit	Pass
T5	Name: Mike Johnson Gender: (Empty) Address: 654 Pine St, Tampa, FL DOB: 1999-11-30	Unable to submit	Unable to submit	Pass

Table 1: Data user inputs testing

2. Verification Process Testing

With the information from T1 stored in the network, now we test different verification requests.

Test #	Verification Request	Expected return value	Results	Pass/Fail
T1	{ "Type": "AND", "Criteria": [{ "Type": "Rule", "Field": "Name", "Value": "Santiago Delgado", "Operation": "equals" }] }	True	True	Pass
T2	{ "Type": "AND", "Criteria": [{ "Type": "Rule", "Field": "Gender", "Value": "M", "Operation": "equals" }] }	True	True	Pass
T3	{ "Type": "AND", "Criteria": [{ "Type": "Rule", "Field": "Address", "Value": "Fake Address 123", }] }	True	True	Pass

	WPB, FL", "Operation": "equals" }] }			
T4	{ "Type": "AND", "Criteria": [{ "Type": "Rule", "Field": "Age", "Value": "22", "Operation": "equals" }] }	True	True	Pass
T5	{ "Type": "AND", "Criteria": [{ "Type": "Rule", "Field": "Age", "Value": "18", "Operation": "minimum" }] }	True	True	Pass
T6	{ "Type": "AND", "Criteria": [{ "Type": "Rule", "Field": "Age", "Value": "30", "Operation": "maximum" }] }	True	True	Pass
T7	{ "Type": "AND", "Criteria": [{ "Type": "Rule", "Field": "Name", "Value": "Santiago Delgado", "Operation": "equals" }, { "Type": "Rule", "Field": "Gender", "Value": "M", "Operation": "equals" }] }	True	True	Pass

	<pre> }, { "Type": "Rule", "Field": "Age", "Value": "22", "Operation": "equals" }, { "Type": "Rule", "Field": "Address", "Value": "Fake Address 123, WPB, FL", "Operation": "equals" }] }</pre>			
T8	<pre> { "Type": "AND", "Criteria": [{ "Type": "OR", "Criteria": [{ "Type": "Rule", "Field": "Name", "Value": "Santiago Delgado", "Operation": "equals" }, { "Type": "Rule", "Field": "Age", "Value": "23", "Operation": "minimum" }] }] }</pre>	True	True	Pass
T9	<pre> { "Type": "AND", "Criteria": [{ "Type": "OR", "Criteria": [{ "Type": "Rule", "Field": "Gender", "Value": "F", "Operation": "equals" }, { "Type": "Rule", "Field": "Age", "Value": "23", </pre>	False	False	Pass

	<pre> "Operation": "minimum" }] }] } </pre>			
T10	<pre> { "Type": "AND", "Criteria": [{ "Type": "NOT", "Criteria": { "Type": "Rule", "Field": "Gender", "Value": "M", "Operation": "equals" } }] } </pre>	False	False	Pass

Table 2: Verification requests testing

Type of Testing	Number of testcases	Test cases passed	Percentage
Information upload	5	5	100%
Verification requests	10	10	100%
Totals	15	15	100%

Table 3: Testing Summary

F. Discussions on Lessons Learned – Santiago Delgado

Throughout the development of this project, several important lessons emerged across both technical and design aspects. One of the most significant insights was the challenge of balancing security with usability. While implementing encryption, key fragmentation, and decentralized storage improves privacy, it also introduces additional complexity in system design and communication between components. Ensuring that these mechanisms operate seamlessly without negatively affecting user experience required careful planning and iterative testing.

Another key lesson involved the importance of system integration. The project combined multiple technologies, including a web interface, backend API, peer-to-

peer networking, and distributed databases. Coordinating these components highlighted the need for clear architecture design and well-defined interfaces. Early testing revealed that even small inconsistencies between modules could lead to failures in data flow or verification processes, emphasizing the value of incremental testing and debugging.

Additionally, the team gained a deeper understanding of the practical limitations of decentralized systems. While decentralization enhances privacy and reduces single points of failure, it also introduces challenges related to scalability, latency, and coordination among nodes. These trade-offs reinforced the decision to adopt a hybrid approach, combining decentralized storage with a trusted authority for initial verification, which proved to be more feasible within the project scope.

Finally, collaboration and task distribution played a crucial role in the project's success. Working as a team required clear communication, division of responsibilities, and consistent progress tracking. This experience highlighted the importance of teamwork in engineering projects, especially when dealing with complex, multidisciplinary systems.

IV. CONCLUSION – Santiago Delgado

In this project, we successfully designed and implemented a proof-of-concept decentralized storage network for online identity verification. The system demonstrates that it is possible to verify user identity attributes without exposing sensitive personal information, addressing a critical issue in modern digital services. By combining encryption, key fragmentation, and a peer-to-peer network, the solution reduces reliance on centralized databases and minimizes the risk of large-scale data breaches.

The proposed approach achieves a practical balance between privacy, security, and usability. Unlike traditional methods that require full disclosure of personal data, our system allows verifiers to receive only a confirmation response, preserving user anonymity while maintaining trust in the verification process. The inclusion of a one-time trusted authority verification step further strengthens the system's reliability without compromising its decentralized nature.

Testing results confirmed that the system performs as expected, successfully handling data input validation and a wide range of verification requests. These results support the feasibility of the design and demonstrate its potential for real-world applications, particularly in scenarios where privacy-sensitive identity verification is required.

If given additional time, future improvements could focus on enhancing scalability, optimizing network performance, and integrating more advanced cryptographic techniques such as full zero-knowledge proofs. Expanding the system to operate in a larger, real-world distributed environment would also provide further validation of its effectiveness.

Overall, this project contributes to the ongoing effort to develop secure, privacy-preserving identity verification systems and demonstrates a viable direction for future research and implementation.

V. REFERENCES

- Tyagi, A., Sahoo, S. K., Katarmal, L., Gupta, M., & Luthra, L. (2018). Online identity verification. *International Journal of Recent Research Aspects*, 5(1), 133–137.
- Satybaldy, A., Subedi, A., & Nowostawski, M. (2022). A framework for online document verification using self-sovereign identity technology. *Sensors*, 22(21), 8408. <https://doi.org/10.3390/s22218408>
- Forland, S., Meysenburg, N., & Solis, E. (2024). Age verification: The complicated effort to protect youth online. Open Technology Institute.
- Eltaher, F., Gajula, R., Miralles-Pechuán, L., Thorpe, C., & McKeever, S. (2025). The digital loophole: Evaluating the effectiveness of child age verification methods on social media. In *Proceedings of the School of Computer Science Conference*. Technological University Dublin. <https://arrow.tudublin.ie/scschcomcon/442>
- Jarvie, C., & Renaud, K. (2021). Are you over 18? A Snapshot of Current Age Verification Mechanisms. *Proceedings of 2021 IFIP 8.11/11.13 Dewald Roode Information Security Research Workshop*, San Antonio, Texas, USA.
- Marsden, C. (2023). Age-Verification Laws in the Era of Digital Privacy. *National Security Law Journal*, 10(2), 210-244.
- Alizadeh, M., Andersson, K., & Schelén, O. (2022). Comparative analysis of decentralized identity approaches. *IEEE Internet of Things Journal*. Retrieved from https://www.researchgate.net/publication/363104774_Comparative_Analysis_of_Decentralized_Identity_Approaches
- Takeuchi, K., Kumamoto, T., Yoshida, Y., & Fujima, H. (2023). Decentralized identity verification system for data access to prevent data exfiltration ransomware. Retrieved from https://www.researchgate.net/publication/376306816_Decentralized_Identity_Verification_System_for_Data_Access_to_Prevent_Data_Exfiltration_Ransomware
- Jamal, A., Syahirah, A. S. N., Helmi, R. A. A., & Fatima, M.-A. F. (n.d.). Blockchain-based identity verification system. *IEEE*. Retrieved November 16, 2025, from <https://ieeexplore.ieee.org/document/8906403>
- Deng, H., Liang, J., Zhang, C., Liu, X., Zhu, L., & Guo, S. (2024). FutureDID: A fully decentralized identity system with multi-party verification. *IEEE Transactions on Computers*, 73(8), 2051–2068. <https://doi.org/10.1109/TC.2024.3398509>
- Dib, O., & Toumi, K. (2020). Decentralized identity systems: Architecture, challenges, solutions and future directions. *Annals of Emerging Technologies in Computing*, 4(5), 19–40
- Bachmann, A., Barbir, A., Brennan, J., Dingle, P., Avellaneda, O., Maler, E., Reed, D., Sporny, M., & Duffy, K. H. (2019). Decentralized identity: Where did it come from and where is it going? *IEEE Communications Standards Magazine*, 3(4), 10–13